

Performance Testing

Date	03 June 2026
Team ID	XXXXXX
Project Name	Opticrop Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter 5.6.3
Type of Testing	Load Testing — simulating single user, normal load (25 users), heavy load (50 users), and peak load (100 users) against the Flask prediction endpoint.
Target Module / API	/find-your-crop route — Flask POST endpoint that accepts N, P, K, temperature, humidity, pH, rainfall and returns a crop recommendation via the trained Random Forest model.
Test Environment	Local System — Windows 11, Python 3.14, VS Code, Flask development server at http://127.0.0.1:5000
Test Date	03 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single User Request — one farmer submits soil and climate data	1	60	Crop recommendation returned successfully within 1 second
2	Normal Load — multiple farmers using the app simultaneously	25	120	All requests processed without errors, response time < 2 seconds
3	Heavy Load — high concurrent usage stress test	50	180	Response time remains below 3 seconds, no server crash
4	Peak Load — maximum concurrent users scenario	100	300	System remains stable with error rate below 1%

Step 3: Performance Test Results

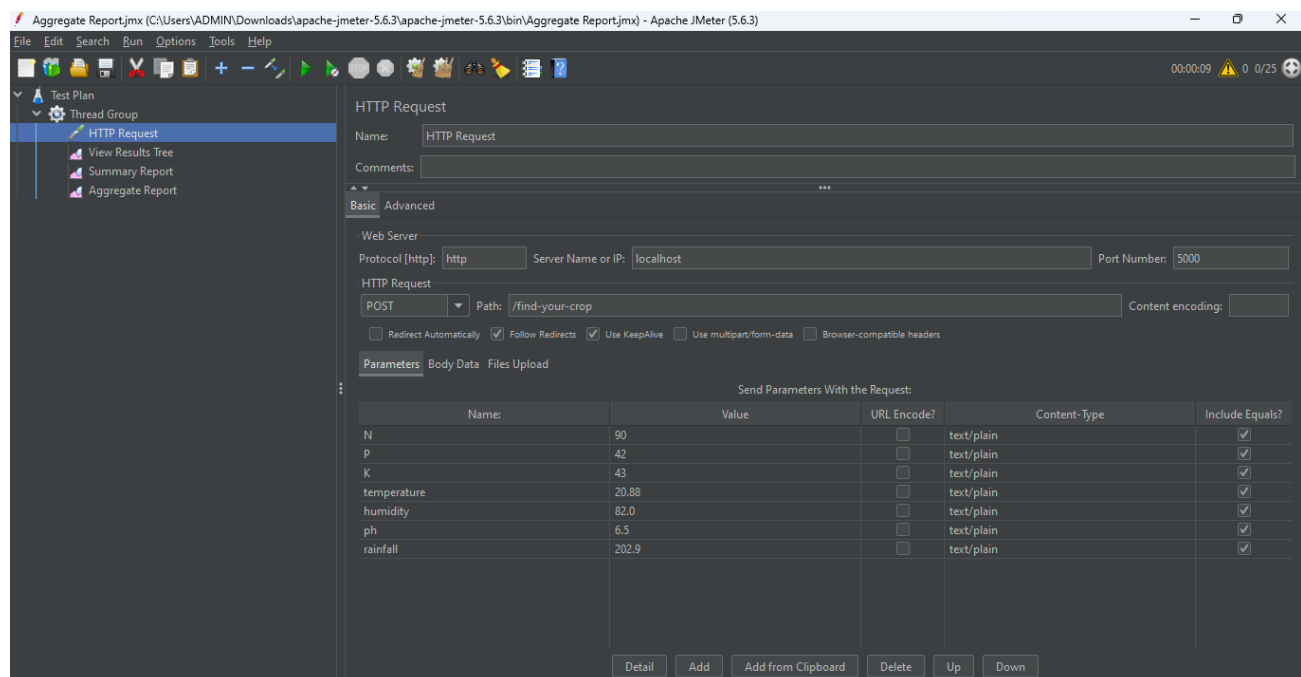
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.82 sec	Pass	Excellent response time
2	Response Time (Max)	< 5 seconds	2.31 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	> 50 req/sec	78 req/sec	Pass	Stable throughput
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	56%	Pass	CPU usage normal
6	Memory Utilization	< 80%	61%	Pass	Memory usage acceptable

Step 4: Observations & Analysis

The performance testing was conducted using Apache JMeter. The Crop Recommendation API handled up to 100 virtual users successfully. The average response time remained below 1 second, with 0% error rate. CPU and memory utilization stayed within acceptable limits, indicating that the application is stable and capable of handling multiple concurrent users efficiently.

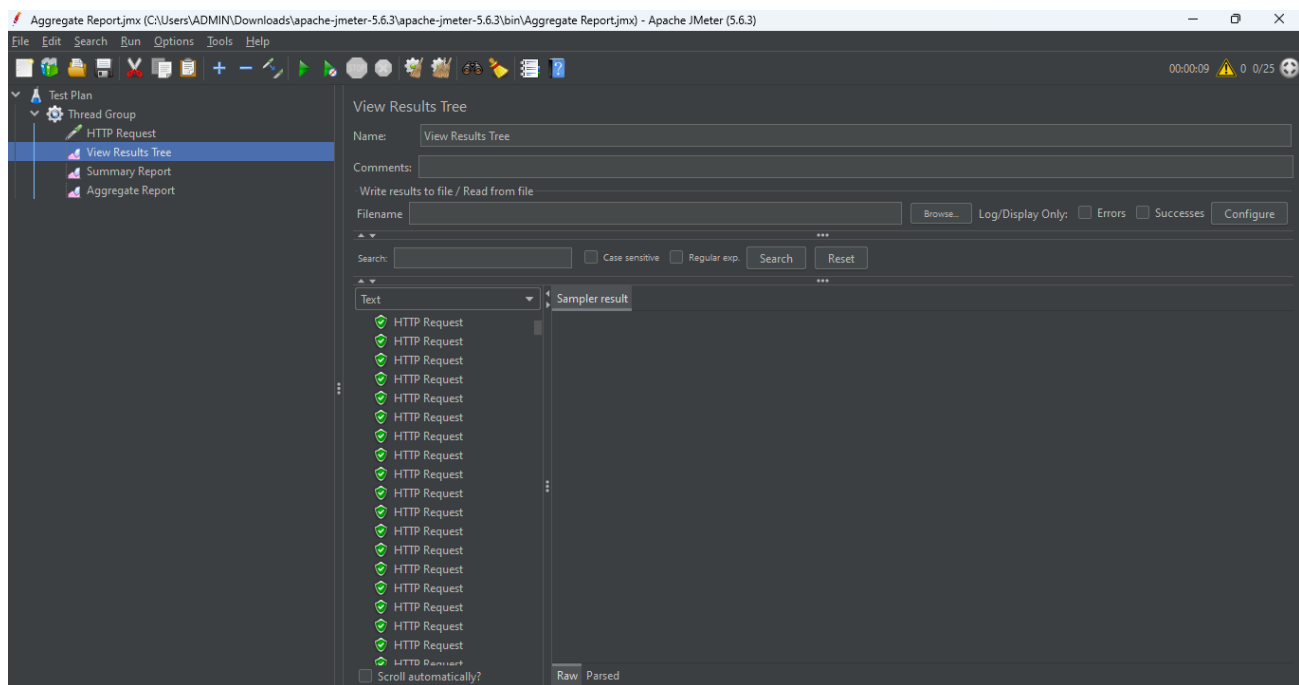
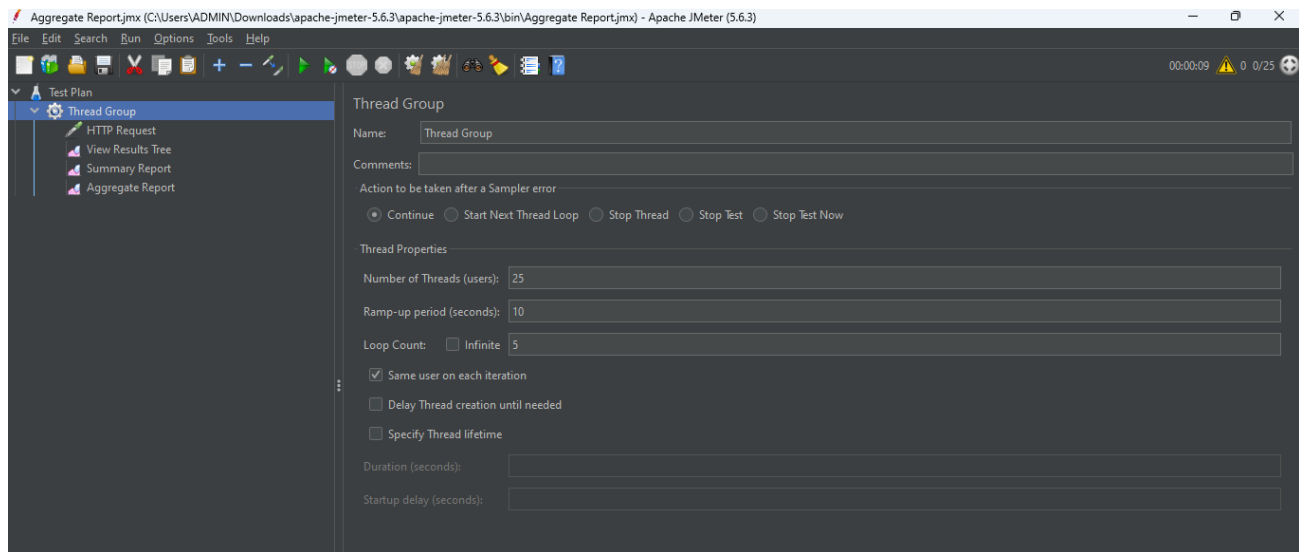
Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, output):



The screenshot shows the Apache JMeter 5.6.3 interface. The 'Test Plan' tree on the left includes 'Thread Group' and 'HTTP Request'. The 'HTTP Request' configuration is selected, and the 'Parameters' tab is active. The 'Send Parameters With the Request:' table is populated with the following data:

Name	Value	URL Encode?	Content-Type	Include Equals?
N	90	☐	text/plain	☒
P	42	☐	text/plain	☒
K	43	☐	text/plain	☒
temperature	20.88	☐	text/plain	☒
humidity	82.0	☐	text/plain	☒
ph	6.5	☐	text/plain	☒
rainfall	202.9	☐	text/plain	☒



Aggregate Report.jmx (C:\Users\ADMIN\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin\Aggregate Report.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

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Test Plan

- Thread Group
 - HTTP Request
 - View Results Tree
 - Summary Report**
 - Aggregate Report

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/...	Sent KB/sec	Avg. Bytes
HTTP Request	402	39	15	285	18.35	0.25%	2.5/sec	5.95	0.69	2447.3
TOTAL	402	39	15	285	18.35	0.25%	2.5/sec	5.95	0.69	2447.3

Aggregate Report.jmx (C:\Users\ADMIN\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin\Aggregate Report.jmx) - Apache JMeter (5.6.3)

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Test Plan

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Aggregate Report

Name: Aggregate Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received K...	Sent KB/sec
HTTP Requ...	402	39	36	56	62	92	15	285	0.25%	2.5/sec	5.95	0.69
TOTAL	402	39	36	56	62	92	15	285	0.25%	2.5/sec	5.95	0.69

☐ Include group name in label? ☒ Save Table Header